Use-Case Flow: Log Maintenance Ticket

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #07 — Hostel Maintenance & Issue Ticketing System

Actors

Primary: Hostel Resident | Supporting: Maintenance Warden (receives the routed ticket)

Preconditions

The resident is logged into the hostel ticketing app/portal with a verified room/block assignment on file.

Main Success Scenario

1. Resident selects “Log New Ticket” from the app home screen. System displays a ticket form with fields for room number tag, issue category, and description.
2. Resident confirms or edits the pre-filled room number tag (auto-detected from their profile).
3. Resident selects an issue category — Plumbing, Electrical, or Carpentry — and a severity level (Low/Medium/High).
4. Resident optionally attaches a photo of the issue (Attach Photo Evidence).
5. Resident submits the ticket.
6. System validates that all required fields (room tag, category, severity) are present.
7. System generates a unique ticket ID and timestamps the submission.
8. System routes the ticket to the Maintenance Warden's queue, sorted by severity.
9. System displays a confirmation screen to the resident showing the ticket ID and expected initial-response time.
10. Use case ends successfully.

Alternate Flow — 6a: Incomplete Submission

Triggered at step 6 if a required field (most commonly the room number tag) is missing or invalid.

a) System rejects the submission and highlights the missing/invalid field(s) in red.
b) System displays a message: “Please complete all required fields before submitting.”
c) Resident corrects the highlighted field(s) and resubmits.

Resumes at step 6 once the resident resubmits; the ticket is not created and no ticket ID is issued until validation passes.

Postconditions

Success: a new ticket exists with a unique ID, is visible in the resident's ticket history, and appears in the warden's queue. | Failure: no ticket record is created.